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United States Patent	12405131
Kind Code	B2
Date of Patent	September 02, 2025
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Terrain verification system and method for pallet drop

Abstract

A work machine includes a terrain verification system for dropping a pallet on a land surface at a destination. Operating autonomously or semi-autonomously, the work machine includes a range sensor, an image sensor, and an inertial sensor to assess the slope and roughness of the land surface on approach to the destination. The terrain verification system generates a ground map of the destination from range data received by the range sensor and generates a roughness metric indicative of the surface. Image data received by the image sensor is used to verify the roughness metric and, when obstructions are found from the image data, the system moves the work machine over the land surface and evaluates pitch and roll data from the inertial sensor to determine the condition of the destination for dropping the pallet.

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Appl. No.: 18/115489

Filed: February 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240288280 A1	Aug. 29, 2024

Publication Classification

Int. Cl.: G01C21/00 (20060101); B65G67/24 (20060101); G01B21/30 (20060101); G01C11/02 (20060101); G01S17/89 (20200101); G05D1/00 (20240101)

U.S. Cl.:

CPC **G01C21/3826** (20200801); **B65G67/24** (20130101); **G01B21/30** (20130101);
G05D1/0214 (20130101); G01C11/02 (20130101); G01S17/89 (20130101)

Field of Classification Search

CPC: G01C (21/3826); G01C (11/02); B65G (67/24); G01B (21/30); G05D (1/0214); G01S (17/89)

USPC: 701/23

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a work machine having a terrain verification system and a method for dropping a pallet on the terrain. More specifically, the present disclosure relates to a work machine and terrain verification system that use an image sensor, a range sensor, and an inertial sensor to ensure a pallet is dropped by the work machine on smooth terrain.

BACKGROUND

(2) Work machines, such as compact track loaders or fork lifts, may travel around a worksite to perform various tasks, such as delivering material to one or more locations or drop sites within the worksite. While traditionally operated manually, these work machines may also be partially controlled by software on the work machine and/or a central computing system, i.e., semi-autonomous operation, or completely managed by software without operator involvement, i.e., autonomous operation.

(3) In either mode of operation with computer control, challenges exist in determining whether the location, or drop site, is safe for delivery of the material. For example, delivery routes for the work machines may span a large outdoor area. While the material may require a relatively flat or level surface on which it may be placed, the ground at the drop site may be uneven due to sloping, troughs, or ruts, or may contain obstructions such as rocks, debris, or vegetation. Unaware of this unevenness, an autonomous or semi-autonomous work machine programmed to deliver the material to the drop site may release the load onto the surface, possibly leading to damage of the material.

(4) Various systems have been developed for using sensors to affect operation of work machines around a job site. These systems, however, do not consider using multiple sensors in combination to evaluate the condition of a drop site before releasing a payload of material, such as dropping a pallet from forks of the work machine.

(5) One approach for using sensors to assist movement of a work machine at a job site is described in U.S. Pat. No. 10,458,938 ("the '938 patent"). The '938 patent describes an industrial truck, such as those used in warehouses or outdoor storage depots, having a sensor device and a memory. The sensor device records into memory irregularities along a delivery route, such as potholes or bumps. Using the history of irregularities, the truck may automatically influence a driver's control when on the delivery route again by reducing speed or setting a steering angle to minimize the risk in hitting the irregularities or by providing a warning. Among other things, the '938 patent does not contemplate using a plurality of sensors to detect the levelness or smoothness of terrain at a drop site not previously traversed by an autonomous or semi-autonomous work machine. As a result, the '938 patent is not desirable for work machines programmed to release material at a drop site.

(6) Examples of the present disclosure are directed to overcoming deficiencies of such systems.

SUMMARY

(7) In an aspect of the present disclosure, a movable work machine includes a chassis, a range sensor, an image sensor, and an inertial sensor. The range sensor is coupled to the chassis and configured to generate range data indicative of topology of the land surface at a destination of the movable work machine. The image sensor is coupled to the chassis and configured to generate image data indicative of the topology. The inertial sensor, also coupled to the chassis, is configured to generate orientation data for the movable work machine during traversal of the land surface. The moveable work machine further includes a terrain verification system that includes a terrain

mapper, an obstruction detector, a localization system, and a roughness processor. The terrain mapper is configured to convert the range data into a ground map of the land surface, and the obstruction detector is configured to identify one or more obstructions on the land surface from the image data. The localization system is configured to evaluate orientation of the movable work machine from at least the orientation data. A roughness processor is configured to determine a roughness metric representative of the topology of the land surface based on at least ground map and to evaluate the roughness metric for accuracy with respect to the image data.

(8) In another aspect of the present disclosure, a method for verifying terrain at a drop site for a work machine, includes receiving, by the work machine, an image signal indicative of image data of land surface at the drop site, and receiving, by the work machine, a range signal indicative of range data of the land surface. The method further includes detecting, by the work machine, an obstruction on the land surface from the image data, and, based at least in part on the detecting the obstruction, causing the work machine to move over the land surface. During movement over the land surface, the work machine receives inertial data indicative of at least pitch and roll of the work machine. The method continues with determining a roughness metric for the land surface based at least in part on the pitch and roll of the work machine from the inertial data.

(9) In yet another aspect of the present disclosure, a semi-autonomous compact loader includes a chassis, traction devices coupled between the chassis and ground, and a work tool coupled to a front of the chassis with respect to a direction of forward travel of the semi-autonomous compact loader, where the work tool is configured to enable lifting and lowering a payload. The semi-autonomous compact loader also includes a range sensor configured to generate range data indicative of topology of a land surface at a destination proximate to the semi-autonomous compact loader, an image sensor configured to generate image data indicative of the topology, and an inertial sensor configured to generate orientation data for the semi-autonomous compact loader during traversal of the land surface. Further, a terrain mapper is configured to convert the range data into a ground map of the land surface, an obstruction detector is configured to identify one or more obstructions on the land surface from image data, and a localization system is configured to evaluate orientation of the semi-autonomous compact loader from at least the orientation data. A roughness processor is configured to determine a roughness metric representative of the topology of the land surface based on at least the ground map and to evaluate the roughness metric for validity with respect to the image data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a top view schematic diagram of a worksite in accordance with an example of the present disclosure.
- (2) FIG. 2 is a side view of a semi-autonomous work machine and block diagram in accordance with an example of the present disclosure.
- (3) FIG. 3 is a top view of the semi-autonomous work machine of FIG. 2 approaching a destination drop site in accordance with an example of the present disclosure.
- (4) FIG. 4 is a flowchart of a first method of verifying terrain at a destination for delivering cargo in accordance with one example of the present disclosure.
- (5) FIG. 5 is a flowchart of a second method of verifying terrain at a destination for delivering cargo in accordance with another example of the present disclosure.

DETAILED DESCRIPTION

(6) Consistent with the principles of the present disclosure, an autonomous or semi-autonomous work machine may include a terrain verification system, typically embedded within its hardware and/or software, for verifying the condition of a land surface at a drop site for the work machine. In

some examples, the work machine is configured with forks and may carry a payload on the forks for delivery to the drop site. Using sensors on the work machine, the terrain verification system generates a ground map of the land surface at the drop site using collected range data and checks for any obstructions on the land surface using collected image data. In some examples, when image analysis detects an obstruction, the terrain verification system may cause the work machine to drive over the drop site and use inertial data from inertial sensors to assess whether the obstruction is deformable (e.g., vegetation) or would impede the drop operation (e.g., rocks or logs). The following describes several examples for carrying out the principles of this disclosure. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

(7) FIG. 1 is a top view schematic diagram of an example system **100** including a computing system **102** and machines **104** within a worksite **106**. During a job at worksite **106**, a fleet of one or more machines **104** can be configured to deliver material from a pick-up location to a set of drop points **108** within the worksite **106**. Worksite **106** can be any geographic area for distribution, although the present disclosure is directed to exposed, outdoor land. In one non-limiting example, worksite **106** can be a solar farm that is under construction across an expanse of land. The drop points **108** can be locations at worksite **106** where machines **104** are to deliver solar panels, solar panel installation equipment, and/or other materials. In some examples discussed further below, machines **104** may include a work tool in the form of forks for lifting, carrying, and placing the materials packaged with or resting on a pallet.

(8) While discussed for illustration as a solar farm, worksite **106** can be a construction site, a mine site, a quarry, or any other type of worksite or work environment at which material can be delivered to defined drop points **108**. The machines **104** at worksite **106** can be compact track loaders (CTLs), skid steer loaders, track loaders, haul trucks, wheel loaders, or any other type of machine that can transport material from one location to another at worksite **106**. The material can be solar panels, construction materials, dirt, gravel, and/or any other type of material that is to be delivered to drop points **108** at worksite **106**.

(9) Computing system **102** can be one or more servers, computers, or other off-board computing devices that are separate from machines **104**. For example, while machines **104** are located at the worksite **106**, computing system **102** can be located at a back office or other location that is remote from machines **104** at the worksite **106**, or that is remote from worksite **106** overall. As illustrated in FIG. 1, computing system **102** can communicate wirelessly with at least machines **104** via a network **110**. In some examples, computing system **102** can generate machine instructions **112** associated with a job for one or more of machines **104** to operate autonomously or semi-autonomously within worksite **106**.

(10) As discussed above, drop points **108** can be locations within worksite **106** that are to receive a payload of material delivered by machines **104**. Staging area **128** can be an area where machines **104** can pick up or obtain the material, before transporting the material from staging area **128** to one or more individual drop points **108**. For example, if worksite **106** is a solar farm that is under construction, delivery trucks or other machines can deliver a set of solar panels to staging area **128**. One or more machines **104**, such as CTLs under the direction of machine instructions **112**, can then transport individual solar panels from staging area **128** to individual drop points **108** at or near where the solar panels will be installed. In some examples, there may be more than one staging area **128** at worksite **106**.

(11) Paths **130** can be roads, passable routes, and/or other pathways through the worksite **106**. For example, the paths can extend from staging area **128** and drop points **108**, such that machines **104** can navigate paths **130** to deliver material from staging area **128** to drop points **108**. The drop points **108** can be located on or near paths **130**. In some examples, paths **130** can include main paths **132** and/or dead-end paths **134**, as shown in FIG. 1. The dead-end paths **134** may branch off from main paths **132**.

(12) Machine instructions **112** for a particular machine **104**, whether received from computing system **102** or previously stored within the machine, can indicate when and/or where the particular machine is to load material, which drop points **108** the machine is to deliver the material to, route segments that the machine is to traverse in a loaded or unloaded state, when the machine is to traverse such route segments, when and/or where the machine is to wait at worksite **106** while loaded or unloaded, and/or other scheduling or routing information. Machine instructions **128** for a machine can also indicate instructions associated with machine operations that the machine is to perform at locations within worksite **106**, such as particular loading or unloading operations, or speeds the machine is to travel at during one or more segments of a cycle.

(13) Discussed in more detail for FIGS. **2** and **3**, machines **104** can be vehicles or other mobile machines that have engines, motors, drivetrains, braking systems, hydraulic components, and/or other mechanisms that can cause movement of wheels of the machines, movement of work tools and/or other elements of the machines, and/or otherwise implement operations of the machines. In some examples, one or more of the machines **104** can be fuel-powered machines that operate based on power provided by internal combustion engines and/or other elements that consume fuel. In other examples, one or more of the machines **104** can be battery electric machines (BEMs), battery electric vehicles (BEVs), hybrid vehicles, fuel cell and battery hybrid vehicles, or other mobile machines. For instance, machines **104** can have batteries, such as lithium-ion (Li ion) batteries, lithium-ion polymer batteries, nickel-metal hydride (NiMH) batteries, lead-acid batteries, nickel cadmium (Ni—Cd) batteries, zinc-air batteries, sodium-nickel chloride batteries, or other types of batteries that can at least partially power the machines **104**.

(14) Machines **104** can be semi-autonomous machines or fully autonomous machines that operate automatically based on machine instructions **112** generated by the computing system **102** or another computing system. For example, a machine can have an electronic control module (ECM) **122** and/or other on-board computing devices that can fully or partially control operations of the machine, such as steering, speed adjustments, work tool movements, and/or other operations. Accordingly, operations of a machine can be fully or partially controlled, automatically or semi-automatically, by on-board controllers such as the ECM **122**, and/or off-board controllers such as the computing system **102**. A machine can, for instance, have an on-board guidance system that can drive the machine autonomously, an obstacle detection system that assists the on board guidance system or can alert a human operator of nearby objects detected by the obstacle detection system, and/or other systems that fully or partially control operations of the machine. As another example, the computing system **102** or another off-board computing device can receive data from a machine and return machine instructions **112** to machine **104** to dispatch the machine to autonomously travel along a defined and/or assigned route, or to fully or partially control operations of the machine remotely.

(15) Machines **104** can also include sensors **124**, such as cameras, LIDAR sensors, RADAR sensors, other optical sensors or perception systems, Global Positioning System (GPS) sensors, other location and/or positioning sensors, work tool position sensors, hydraulic pressure sensors, payload sensors, speed sensors, brake temperature sensors, other temperature sensors, tire pressure sensors, battery state of health (SoH) sensors, fuel sensors, incline and decline travel sensors, and/or other types of sensors. Sensors **124** of a machine can be operable coupled to the ECM **122** and/or other on-board computing systems of the machine. Sensors **124** can also provide corresponding sensor data to the ECM **122** and/or other on-board computing systems of the machine, and/or off-board computing systems such as computing system **102**, such that the sensor data can be used to determine a location of the machine, detect nearby terrain, detect nearby objects, such as vehicles, other machines, or personnel, detect the positions of such nearby objects relative to the machine, determine a weight of a payload carried by the machine, determine a state of charge (SoC) of a battery system, determine an amount of fuel carried by the machine, and/or perform other operations. In some examples, data provided by sensors **124** of a machine can enable

the ECM **122** of the machine to cause the machine to drive and/or operate autonomously or semi-autonomously.

(16) Machines **104** can have wireless communication interfaces **126** that are operably coupled to the ECMs **122** of the machines **104**, and that allow the ECMs of the machines **104** to send data to computing system **102** and/or other off-board controllers, and to receive machine instructions **120** and other data from the computing system **102** and/or other off-board controllers. Such wireless communication interfaces **126** can include cellular interfaces, modems, receivers, transmitters, antennas, and/or other hardware or software elements configured to send and receive data, for instance to exchange data with computing system **102** and/or other off-board controllers. Computing system **102** and/or other off-board controllers can have, or be associated with, similar wireless communication interfaces, such that computing system **102** and/or other off-board controllers can wirelessly exchange data with ECMs **122** and/or other on-board computing systems of the machines **104**.

(17) While FIG. **1** provides context for the operation of machines **104**, FIGS. **2** and **3** together illustrate a semi-autonomous compact track loader (SACTL) as one example of a machine **204** within the machines **104**. FIG. **2** is a side view and functional block diagram **200** of machine **204**. As shown, machine **204** includes a chassis **206** to which is coupled a power source **208** configured to supply power for operation of machine **204**, including, for example, operating work implements, electronics, and steering, and/or for supplying torque to drive members to propel machine **204** across one or more paths **130**. For example, machine **204** shown in FIG. **1** includes traction devices in the form of a pair of tracks **210** coupled to the chassis **206** and configured to propel machine **204** across paths **130**. Alternatively, machine **204** could include one or more wheels instead of, or in addition to, tracks **210**. The example machine **204** also includes a cab **212** coupled to chassis **206** for protecting and/or providing comfort for an operator of machine **204** and/or for protecting control-related devices of machine **204**. In some examples, machine **204** may be fully autonomous and able to operate without an onboard or remote human operator.

(18) Machine **204** may be fitted with a variety of work implements well known in the field, such as a blade, a broom, a shovel, or a plow, for manipulation by arms **214**. In the example of FIG. **2**, machine **204** includes a pair of forks **214** attached to arms **216**. Forks **216** enable machine **204** to function as a forklift, sliding under and lifting material for transportation between staging area **128** and one or more drop points **108**. Referring to FIG. **2**, forks **216** are depicted under a pallet **218** on which a payload, such as cargo **220**, rests. In one example, cargo **220** may be solar panels, solar panel installation equipment, and/or other materials for distribution within worksite **106**. It will be understood that pallet **218** is optional for transporting cargo **220**, and forks **216** may be used to lift, transport, and drop cargo **220** without use of pallet **218**.

(19) Among sensors **124** depicted in FIG. **1** for machines **104**, machine **204** includes at least one image sensor **222**. Image sensor **222** is configured to sense, determine, and/or collect environmental information relating to visual images in one or more perspectives about machine **204** as image data **224** and to generate one or more image signals that include or indicate this environmental image data **224**. Image sensor **222** may be any known type of analogue or digital image sensor, digital camera, and/or digital video camera. For example, image sensor **222** may be a high dynamic range (HDR) camera, a light-sensitive camera, and/or an ultra-sonic camera. Other suitable types of imagers are contemplated. Image sensor **222** captures still or moving image data **224** suitable for use by machine **204** discussed below. In some examples, image sensor **222** provides two-dimensional image data, three-dimensional image data, image sequences, gray image data, and/or color image data. In some examples, image sensor **222** provides depth data, absorption data, and/or reflectance data.

(20) Also among sensors **124** for machines **104**, machine **204** includes at least one range sensor **226**. Range sensor **226** is configured to sense, determine, and/or collect environmental information relating to depths and/or ranges of objects in one or more perspectives about machine **204** as range

data **228**. Further, range sensor **226** generates one or more range signals that include or indicate this range data **228**. Range data **228** may be used for detecting objects and land forms in the environment about machine **204**. For example, range sensor **226** may be a machine “vision” device used in machine navigation, obstacle detection and avoidance, collision detection and avoidance, site mapping, material excavation or extraction, and/or applications. Accordingly, range sensor **102** may be a device configured to determine the range and direction from range sensor **226** to points on a surface within a field of view of range sensor **226**. In some examples, as discussed further below, range sensor **226** collects range data **228** regarding the contour of terrain in front of forks **216** in the direction of forward travel for machine **204** for generation of a terrain map. In one example, range sensor **226** includes or is part of a Light Detection and Ranging (LIDAR) device. In another example, range sensor **226** includes or is part of stereo cameras configured for providing depth assessment.

(21) When implemented as a LIDAR unit, range sensor **226** may include a plurality of light sources, such as lasers. Each laser generates a laser beam which is directed at various points of worksite **106**. In this example, range sensor **226** further includes one or more detector devices that receive the laser beams after reflection off of various points of worksite **106**. Based on the time between generating the laser beam and receiving the reflected laser beam, range data **228** determines a distance to the corresponding point. As a result, range sensor **226** and additional processing discussed below will generate a 3D point cloud image or grid map representative of a part of worksite **106** that is detected by the LIDAR unit and is part of range data **228**.

(22) Also among sensors **124** for machines **104**, machine **204** includes at least one inertial sensor **230** and one or more proprioceptive sensors (not shown). The at least one inertial sensor **230** is configured to detect changes in position and orientation during movement of machine **204** as inertial data **232**. In some examples, inertial data **232** arises from an inertial sensor **230** that includes three orthogonally oriented accelerometers for detecting changes along an X-axis, Y-axis, or Z-axis, and three orthogonally oriented gyroscopes for detecting changes in pitch rate, yaw rate, and roll rate during movement of machine **204**. In other examples, inertial sensor **230** also includes three orthogonally oriented magnetometers serving as a three-dimensional electronic compass and/or a two-axis inclinometer to provide pitch angle and roll angle. Inertial sensor **230** is typically part of an inertial measurement unit (IMU) within machine **204**. The one or more proprioceptive sensors detect the linear movement of machine **204**, such as the speed of tracks **210** (or wheels), which may be used to determine the velocity of machine **204**.

(23) As also shown in FIG. 2, functional block diagram **200** for machine **204** includes a terrain verification system **234** for evaluating the quality of terrain within one of the drop points **108** in which cargo **220** is to be deposited. Cargo **220** often requires that the surface on which it is deposited by machine **204** is relatively level and flat. In some examples, cargo **220** may include electronic equipment, glass panels, or other material subject to breakage, and depositing cargo **220** within one of drop points **108** that is uneven or contains obstructions can damage cargo **220** or lead to installation or usage problems. Moreover, unevenness could cause forks **216** to drag or catch on the ground and disrupt a proper delivery. While the relevant drop point may have been prepared or graded previously, when the drop point is outdoors, weather or other factors may have deteriorated the condition of the land surface. For instance, rain may have created ruts at the intended drop point, vehicle tracks may have caused an unevenness to the surface, or rocks may be present. Further, grass, bushes, or other vegetation may have grown at the location. Terrain verification system **206** enables machine **204** to determine independently through sensor collection and computer processing whether the land surface of the intended one of drop points **108** for cargo **220** is within an acceptable condition for delivery.

(24) Terrain verification system **234**, which is illustrated in FIG. 2 as a functional block for machine **204**, is a system within the work machine for receiving image data **224**, range data **228**, and inertial data **232** from image sensor **222**, range sensor **226**, and inertial sensor **230**,

respectively, and evaluating the sufficiency of the topology at one of the drop sites **108** for delivering a payload. Described in FIG. **2** and this disclosure with respect to functional blocks, terrain verification system may be incorporated as hardware, software, or a combination of hardware and software within control electronics of machine **204**. For instance, in one implementation, terrain verification system **234** may be embodied within hardware in the form of a supplemental ECM, configured to execute computer code to perform the detection, analysis, and instructions relating to verifying the terrain at one of the drop sites **108**. In this example, the supplemental ECM (not shown) would interface with existing ECM **122** as well as with appropriate detection modules within machine **204**, such as image sensor **222**, range sensor **226**, and inertial sensor **230** to help coordinate the activities for verifying the suitability of terrain for dropping cargo **220** at one of the drop sites **108** consistent with the following discussion.

(25) In one example, terrain verification system **234** evaluates the sufficiency of a land surface at one of the drop sites **108** for delivering a payload based on the roughness of the surface.

“Roughness” in this context generally refers to a degree in which the surface at the drop point deviates from a flat plane. The deviation may arise from erosion or other forces causing troughs or other gaps within the land surface at the drop point. Or the deviation may arise from sporadic accumulation of land above the surface in the form of mounds or rocks. Similarly, the deviation from a flat plane may arise from a slope leading to an overall unlevel surface. The degree or quantification of roughness will depend on the implementation and is within the discretion of those of ordinary skill in the art.

(26) Among other functions, terrain verification system **234** stores a predetermined roughness threshold **236** in a memory of machine **204**. Roughness threshold **236** operates as a base line value of a maximum roughness, i.e., deviation from a flat plane, acceptable for the surface at a drop point for delivery of cargo **220**. Roughness threshold **236** may take any form suitable to the implementation, but in one example roughness threshold **236** is a numerical or mathematical representation calculated through statistical methods or modeling of a variety of landscapes and taking into account the characteristics of cargo **220** to be dropped.

(27) Factors affecting a value for roughness threshold **236** may include a slope amount and direction of the surface at a drop point, a variation in the surface from a plane due to mounds or troughs, and the presence of obstructions such as rocks or vegetation. In some examples, roughness threshold **236** is previously calculated from test runs or modeling and, as mentioned, may take into account the type of material for cargo **220** and its sensitivity to breakage and the characteristics of machine **204** and its forks **216**. As discussed below, roughness threshold **236** serves as a benchmark or base line in evaluating the condition of one of the drop sites **108** for smoothness sufficient to effectively and safely carry out the delivery of specific cargo **220**. In general, a higher value for roughness threshold **236** indicates a rougher terrain, i.e., a surface with greater deviation compared with a flat plane, although in other implementations a higher value could alternatively designate a smoother terrain.

(28) Referring to FIG. **3**, context for terrain verification system **234** is shown for a drop approach **300** for machine **204**. FIG. **3** is a top view of machine **204** with cargo **220** approaching a destination **308** consistent with the principles of the present disclosure. In some examples, destination **308** is one of the drop points **108** in FIG. **1** provided in machine instructions **112** for machine **204** to deliver cargo **220** from staging area **128** after traversing one or more of paths **130**. As illustrated in FIG. **3**, destination **308** is in front of forks **216** in a direction of forward travel for machine **204**. In this position, with machine **204** either stationary or moving toward destination **308**, image sensor **222** may receive image information **310** and range sensor **226** may receive range information **312** within a zone including destination **308**. Image information **310** and range information **312** may be collected simultaneously or separately as desired for the intended implementation.

(29) Within destination **308** in FIG. **3**, several potential anomalies to a level and smooth surface for

cargo 220 are shown. For example, destination 308 may include uneven ground 302 possibly due to erosion from weather forming ruts or valleys within dirt. In other situations, uneven ground 302 is caused by elevated portions of earth such as from mounds of dirt above the surface. Also illustrated in FIG. 3, destination 308 has vegetation in the form of grass 304 and bushes 306 that have grown. As discussed below, terrain verification system 234 takes into account and evaluates the presence of uneven ground 302, grass and similar light vegetation 304, and bushes and similar heavy vegetation 306, as well as any other obstacles to a level surface such as rocks or debris, as machine 204 approaches destination 308.

(30) To accomplish the evaluation of destination 308, terrain verification system 234 includes a terrain mapper 238 configured to prepare a map of the terrain at destination 308, as indicated in FIG. 2. Typically implemented as software routines such as within ECM 122, a supplemental ECM (not shown), or other computing electronics in machine 204, terrain mapper 238 assesses depth or range data 228 collected on machine 204 for destination 308 to generate a representation of the land surface at that location. For example, as machine 204 approaches destination 308 to deliver cargo 220, range sensor 226 in the form of a LIDAR unit will generate laser beams as range information 312 across the surface of destination 308 and receive reflected laser beams in response, which are processed as part of depth or range data 228. In some examples, terrain mapper 238 filters out obstacles detected in the environment, such as piles, vehicles, construction debris, personnel, etc., from range data 228. The resulting range data 228 will then represent the landscape of the ground, which may be processed for a ground map. Based in part on the time differences between the transmitted and received beams, terrain mapper 238 can determine distances from a reflected object, e.g., uneven ground 302. As the reflected beams are accumulated, terrain mapper 238 generates a 3D point cloud image, a grid map, or other representation of the surface of destination 308, in a manner known to those of ordinary skill in the field.

(31) In some examples, terrain verification system 234 processes the results of the ground map generated by terrain mapper 238 via a roughness processor 243. Roughness processor 243 is a functional component of terrain verification system 234 that may be implemented within hardware, such as ECM 122, a supplemental ECM, or other computer electronics within machine 204, or software executing on that hardware. Analyzing the ground map generated by terrain mapper 238, roughness processor 243 in general analyzes range data 228 from range sensor 226 to evaluate the condition of the surface at destination 308. Roughness processor 243 calculates a roughness metric 244 as any mathematical assessment to quantify a roughness of destination 308 in terms of the slope, evenness, smoothness, and flatness of the surface. In some examples, a slope value for the surface may be assessed separately from surface roughness or smoothness rather than embedded within roughness metric 244. In some examples, roughness processor 243 calculates roughness metric 244 based on a variance of sensor points within range data 228 collected by a LIDAR or stereo cameras (i.e., range sensor 226) along a vector normal to the ground surface as part of a terrain mapping system. In some examples, roughness metric 244 extends from a low value commensurate with the surface being planar and flat without slope or obstructions to a high value commensurate with the surface being highly disturbed with a steep slope or with large obstructions. The parameters for roughness metric 244 may vary based on the particular implementation and are within the knowledge and experimentation of those skilled in the field.

(32) As also indicated in FIG. 2, in some examples, terrain verification system 234 includes an obstruction detector 240 configured to locate any obstructions for forks 216 and cargo 220 within destination 308. Obstruction detector 240 includes image sensor 222, possibly in coordination with range sensor 226, and associated electronics and software within terrain verification system 234 for processing at least the received image data 224. As machine 204 approaches destination 308 to deliver cargo 220 image sensor 222 will capture image information 310 indicative of image data 224 of the topology of destination 308, either simultaneously with range sensor 226 or separately. Obstruction detector 240 will process image data 224 to determine whether any obstacles exist on

or above the surface of destination **308**. The obstacles may include, for instance, light vegetation **304**, heavy vegetation **306**, or other structures such as rocks or debris. In some examples, obstruction detector **240** processes image data **224** using known image analysis techniques (e.g., known digital analysis techniques), such as by analyzing gradients in the image data **224** to identify obstacles. Deep learning, or machine learning based on artificial neural networks, may also be employed to improve or assist with the image analysis in identifying obstacles, as may other techniques known to those of ordinary skill in the field.

(33) In some examples, components within terrain verification system **234**, such as roughness processor **243**, process the results of the image processing for image data **224** by obstruction detector **240** as a check on the validity of roughness metric **244** as calculated from range data **228** from range sensor **226**. Thus, to the extent range sensor **226** contributes to a map of the terrain generated by terrain mapper **238** that leads to a high value for roughness metric **244**, terrain verification system **234** can use obstruction detector **240** to confirm whether roughness metric **244** is valid or accurate. For instance, obstruction detector **240** may indicate the presence of debris at destination **308**, which may detract from confidence in range data **228** as an indication of ground roughness. Terrain verification system **234** may then take alternative steps to assess ground roughness and roughness metric **244** determined from range data **228** as discussed below.

(34) In other examples, components within terrain verification system **234**, such as roughness processor **243**, process the results of the image processing for image data **224** by obstruction detector **240** as part of calculating roughness metric **244**. Accordingly, in addition to or independent of the ground map processed by terrain mapper **238**, roughness metric **244** in some examples considers results from obstruction detector **240** with respect to visually detectable information received by image sensor **222**. In some examples, image data **224** will influence roughness metric **244** to a higher value when terrain verification system **234** detects one or more obstacles at destination **308** and will not affect roughness metric **244** when no obstructions are detected, although the parameters for roughness metric **244** may vary based on the particular implementation.

(35) Terrain verification system **234** also includes a localization system **242**, as shown in FIG. 2. Localization system **242** is configured to evaluate position and orientation in space of machine **204**. Localization system **242** includes, among other things, one or more components including an IMU that detects inertial data **232** indicative of changes in orientation of machine **204** as tracks **210** move machine **204**. Additionally, localization system **242** may include proprioceptive sensors to detect the speed of tracks **210**, which may be used to determine the velocity of machine **204**. In particular, three orthogonally oriented gyroscopes in an example for inertial sensor **230** provide inertial data **232** including the pitch rate, yaw rate, and roll rate of machine **204**. Accordingly, if machine **204** were to traverse the surface of destination **308**, proprioceptive sensors (not shown) and inertial sensor **230** would provide movement data and inertial data **232** representing the position and orientation of machine **204** including its shifting away from a level condition. Thus, as machine **204** moves over land having deep ruts or heavy debris, localization system **242** will indicate variations in inertial data **232**, such as changes in pitch and roll, representative of a highly uneven surface beneath tracks **210**.

(36) In some examples and situations, terrain verification system **234**, such as roughness processor **243**, takes into account the results of machine movement from localization system **242** in calculating roughness metric **244**. An indication by localization system **242** that the pitch or roll of machine **204**, for example, has changed appreciably while machine **204** traverses destination **308**, may cause terrain verification system **234** to increase roughness metric **244** by a relative amount. In other situations, where little or no change occurs to the pitch or roll of machine **204**, terrain verification system **234** may not affect or may decrease roughness metric **244**, as exemplified below, although the parameters for roughness metric **244** may vary based on the particular implementation.

(37) Consistent with the principles of the present disclosure, terrain verification system 234 considers one or more of image data 224, range data 228, and inertial data 232 in determining the condition of the surface at destination 308, such as by influencing a value for roughness metric 244 by roughness processor 243. Terrain verification system 234 may use the results from one or more of terrain mapper 238, obstruction detector 240, and localization system 242 to verify or cross-check the results from others of terrain mapper 238, obstruction detector 240, and localization system 242. In one example, if the results from terrain mapper 238 leads to an increment for roughness metric 244 by roughness processor 243 due to uneven topology detected through a ground map for destination 308, terrain verification system 234 may consider results from obstruction detector 240 as a check on the results derived from range data 228. Thus, a conclusion from obstruction detector 240 that an obstruction exists in destination 308 would supplement or confirm the result from terrain mapper 238 based on range data 228 of an uneven topology, i.e., a high value for roughness metric 244. Further, in some examples, following a high roughness metric 244 based on input from terrain mapper 238 and obstruction detector 240, terrain verification system 234 causes machine 204 to traverse destination 308 to collect inertial data 232 relevant to the ground surface. If inertial data 232 and image data 224 both indicate that obstructions within destination 308 are light vegetation 304, i.e., obstructions appearing within image data 224 but not affecting pitch or roll of machine 204 in inertial data 232, then terrain verification system 234 may conclude that cargo 220 may be safely deposited at destination. As a result, terrain verification system 234 via roughness processor 243 will decrease roughness metric 224 from its initial value derived from range data 228 sufficiently to permit a release of cargo 220 at destination 308. Other combinations of evaluating image data 224, range data 228, and inertial data 232 are possible, some of which are addressed in more detail in FIGS. 4 and 5 below.

(38) Following calculation and adjustment of roughness metric 244 based on input from one or more of terrain mapper 238, obstruction detector 240, and localization system 242, roughness processor 243 can compare roughness metric 244 to roughness threshold 236. In some examples, if roughness metric 244 is below roughness threshold 236, terrain verification system 234 will allow, or not intervene during, the execution of the drop for cargo 220 at destination 308. In other examples, if roughness metric 244 is equal to or above roughness threshold 236, terrain verification system 234 will interrupt the delivery to protect cargo 220 from potential damage.

(39) As depicted in FIG. 2, terrain verification system 234 is configured to communicate with output device 250. If terrain verification system 234 concludes that either image data 224 or range data 228 needs to be verified with inertial data 232, for example, terrain verification system 234 can provide input to ECM 122, for instance, within output device 250 to cause tracks 210 to traverse machine 204 over destination 308. Similarly, if terrain verification system 234 concludes that the terrain at destination 308 is sufficient for cargo 220, terrain verification system 234 can instruct ECM 122 to complete that mission as provided by machine instructions 112. On the other hand, if roughness processor 243 concludes that the terrain at destination 308 is not sufficient for cargo 220, terrain verification system 234 can interrupt the mission provided by machine instructions 112 and cause output device 250 to communicate via wireless communication interfaces 126 with computing system 102 for further instructions. For semi-autonomous operation, such as with machine 204, output device 250 may alert an operator of the incomplete end of the mission due to the condition of destination 308.

(40) In some examples, terrain verification system 234, including terrain mapper 238, obstruction detector 240, and localization system 242, is implemented in a variety of electronics and software, whether embedded within hardware or separately stored in memory on machine 204. For instance, terrain verification system may include one or more controllers, one or more processors, a memory, and/or other components with respect to machine 204. The controllers may include an electronic control unit, such as ECM 122, a central control module (CCM), a central timing module (CTM), a general electronic module (GEM), a control unit, and/or any other types of ECUs. Other forms of

implementation are within the knowledge of those skilled in the art and are not intended to be limiting to the functional operations described in this disclosure.

(41) Turning from the architecture of machine **204** and terrain verification system **234** as illustrated in FIGS. **2** and **3** to a method involving that architecture, FIG. **4** is a flowchart of a representative method **400** for verifying terrain at a destination **308** for delivering cargo **220** in an example environment, such as a worksite **106**. This process **400** is illustrated as a logical flow graph, operation of which represents a sequence of operations that can be implemented in hardware, software, or a combination thereof. In the context of software, the operations represent computer-executable instructions stored on one or more computer-readable storage media that, when executed by one or more processors, perform the recited operations. Generally, computer-executable instructions include routines, programs, objects, components, data structures, and the like that perform particular functions or implement particular data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described operations may be combined in any order and/or in parallel to implement the process.

(42) In FIG. **4**, the example process **400**, at **402**, includes processing equipment, or other electronic controller or processor on a work machine associated with a terrain verification system, receiving an image signal indicative of image data at a drop point. As shown in FIG. **2**, image sensor **222** positioned on machine **204** captures image information **310** indicative of image data **224** for a drop point, such as destination **308**, which is then provided as an image signal to the controller or processor, such as roughness processor **243**. In some examples, machine **204** captures the image information **310** as machine **204** approaches destination **308** on a mission to deposit cargo **220**. The one or more image signals may be communicated to the processing equipment, such as roughness processor **243**, via a hardwired and/or wireless link for evaluation.

(43) In a second step **404** in FIG. **4**, the processing equipment receives a range signal indicative of depth or range data at the drop point. In one example, range sensor **226**, which may be part of or include LIDAR equipment or stereo cameras as examples, is positioned on machine **204** and receives range information **312** indicative of range data **228** for the drop point, i.e., destination **308**. In some examples, machine **204** captures the range information **312** as machine **204** approaches destination **308** on a mission to deposit cargo **220**. The one or more range signals may be communicated to the processing equipment, such as roughness processor **243**, via a hard-wired and/or wireless link for evaluation.

(44) Step **406** of the method includes the processing equipment calculating a roughness metric based on the range data. In one example, terrain verification system **234** within machine **204** evaluates the content of range data **228** using a terrain mapper **238** to generate a ground map of the earth surface at destination **308**. As discussed above, in one aspect of that activity, terrain mapper **238** may filter out obstacles detected in the environment to yield range data representing only the landscape of the earth surface at destination **308**. Based on this range data **228** and mapping, terrain verification system **234** via roughness processor **243** may prepare or adjust a roughness metric **244** representative of a perceived topology at destination **308**. In more detail, roughness processor **243** may calculate roughness metric **244** based on a variance of sensor points within range data **228** collected along a vector normal to the ground surface as part of the ground map from terrain mapper **238**. If the ground map indicates a surface of varying height, depth, or slope, roughness metric **244** may be increased in value or otherwise adjusted according to the particular implementation.

(45) In step **408** of method **400**, a determination is made whether the roughness metric is above a threshold. As discussed above, a threshold, such as roughness threshold **236** is predetermined in some examples and stored within machine **204**. Whether a numerical value or a more complicated mathematical representation of surface condition, roughness metric **244** may be compared with roughness threshold **236** to determine which is greater. If roughness metric **244** is less than roughness threshold **236**, then the evaluation of terrain from range data **228** indicates that the

surface is within a predetermined acceptable level of smoothness for dropping cargo **220**. As a result (Step **408**—No), the method proceeds to step **410** in FIG. **4**, where the cargo may be released at the drop point, i.e., destination **308**. But if roughness metric **244** is greater than or equal to roughness threshold **236**, then the evaluation of terrain from range data **228** indicates that the surface is not within an acceptable level of smoothness for dropping cargo **220** (Step **408**—Yes), and method **400** proceeds to step **412** for further processing.

(46) As depicted in FIG. **4**, step **412** entails the processing equipment evaluating image data for evidence of an obstruction. In one example discussed above, image data **224** captured by image sensor **222** may be evaluated by obstruction detector **240** through image processing techniques to locate one or more obstructions on the surface of destination **308**. These obstructions may take any form, such as light vegetation **304**, heavy vegetation **306**, rocks, or other debris. If an obstruction is not identified, method **400** at step **414** (Step: **414**—No) interrupts its mission of delivering cargo **220** and at step **416** alerts an operator for guidance. In this instance, a conclusion is reached that range data **228**, such as of uneven ground **302** from LIDAR equipment, accurately reflects the surface conditions at destination **308**, which are not of sufficient quality for delivery of cargo **220**.

(47) On the other hand, if the analysis of image data **224** does identify an obstruction in method step **414** (Step: **414**—Yes), terrain verification system **234** in some examples determines that a high value for roughness metric **244** may be caused by surface debris. Particularly if surface debris is in the form of light vegetation **304** or even heavy vegetation **306**, a LIDAR system as part of range sensor **226** and terrain mapper **238** may provide data leading to the conclusion by roughness processor **243** that a surface is unacceptably rough although the debris may not be an impediment to safe delivery of cargo **220**.

(48) Therefore, following a positive detection of an obstruction in step **414** (Step: **414**—Yes), processing equipment causes the machine to drive over the drop point in step **418** and to receive inertial signals indicative of orientation data in step **420**. As discussed above, terrain verification system **234** in some examples causes machine **204** to drive over destination **308**, during which inertial data **232**, or orientation data, such as pitch and roll of machine **204** is collected. It will be appreciated that if debris located in step **414** has a high degree of deformability, such as due to having a low mass or density, as with light vegetation **304** or certain types of heavy vegetation **306**, tracks **210** of machine **204** will roll over the debris and cause the debris to be crushed or otherwise deformed. As a result, machine **204** will not register significant changes in inertial data **232** due to the deformation of the debris and the comparatively high weight of machine **204**. On the other hand, if debris identified in step **414** has a low degree of deformability, such as due to having a high mass or rigidity and being elevated the ground surface, as with rocks or logs, tracks **210** and the weight of machine **204** will not crush or otherwise deform the debris. Consequently, machine **204** will register changes in its orientation while moving over destination **308**, which will be reflected in movement data and inertial data **232** as detected by localization system **242**.

(49) Following receipt of inertial signals, method **400** in FIG. **4** proceeds to step **422** where processing equipment again calculates the roughness metric based on the orientation data. As noted above, if debris on the ground surface has a low deformation capacity, machine **204** may experience a change in pitch or roll when traversing destination **308**. In this situation, roughness processor **243** will process orientation data or inertial data **232** from localization system **242** and increase roughness metric **244** accordingly. An amount of change to the orientation of machine **204** while traversing destination **308** justifying a change in roughness metric **244** will depend on the particular implementation, including the type of cargo **220** and other factors, and may be determined by the ordinarily skilled artisan. On the other hand, if machine **204** does not experience a significant change in pitch or roll during the traversing, then roughness processor **243** within terrain verification system **234** may decrease roughness metric **244** or keep roughness metric **244** unchanged depending on the circumstances.

(50) As indicated in FIG. **4**, after step **422**, processing equipment such as roughness processor **243**

compares roughness metric **244** to roughness threshold **236** again at step **424**. If, following additional data from obstruction detector **240**, roughness metric **244** is still equal to or above roughness threshold **236** (Step **424**—Yes), method **400** moves to step **416** where the processing equipment causes an alert to be sent to an operator and to stop the delivery mission. In this situation, it will be evident that the surface debris identified by obstruction detector **240** (and perhaps in addition to soil unevenness identified by terrain mapper **238**) is an impediment to the safe delivery of cargo **220**. If, following additional data from obstruction detector **240** and localization system **242**, roughness metric **244** is now below roughness threshold **236** (Step **424**—No), method **400** moves to step **410** where processing equipment will cause cargo **220** to be released at destination **308**. In this situation, it will be evident that roughness processor **243**, in coordination with at least obstruction detector **240** and localization system **242**, concluded that the surface that terrain mapper **238** identified as not a flat plane is of low concern with respect to the delivery of cargo **220** at destination **308**. For example, light vegetation **304** within destination **308** may initially lead to a high roughness metric **244** based on range data **228**, but further analysis using image data **224** and inertial data **232** may result in a lower roughness metric **244** indicating that destination **308** is suitable for dropping pallet **218**. Otherwise, if following analysis image data **224** and inertial data **232** from obstruction detector **240** and localization system **242**, roughness metric **244** is above roughness threshold **236** (Step **424**—Yes), method **400** moves to step **416** where an operator is alerted for guidance.

(51) FIG. 5 illustrates a flowchart of another representative method **500** for verifying terrain at a destination **308** for delivering cargo **220** in an example environment, such as a worksite **106**. Method **500** includes many of the same steps as method **400** but in a different sequence. As a result, the following does not repeat the explanation of those steps but explains the difference in sequence.

(52) As indicated in FIG. 5, an alternative method **500** includes the same opening steps **402** and **404** as in method **400** of processing equipment receiving image signals and receiving range signals. Following this receipt, in method **500** the processing equipment evaluates image data for evidence of an obstruction in step **412** and a decision is made whether an obstruction is detected in step **414**. Thus, different than method **400**, exemplary method **500** first investigates whether visual evidence via image data **224** exists to show an obstruction on the surface of destination **308**. Method **500** takes one of two paths depending on whether evidence of an obstruction is found from image data **224** in step **414**.

(53) If an obstruction is located in image data **224** in step **414**, the processing equipment causes machine **204** to drive over the drop point at destination **308** (step **418**) to receive inertial signals indicative of orientation data, or inertial data **232** (step **420**). Thereafter, in step **422**, roughness processor **243** calculates or updates roughness metric **244** based on the inertial data **232**. As discussed above, if the pitch or roll of machine **204** is substantial as determined for the particular implementation, then roughness processor **243** may increase roughness metric **244**.

(54) On the other hand, if an obstruction is not located in image data **224** in step **414**, terrain verification system **234** will calculate or update roughness metric **244** via roughness processor **243** based on LIDAR or range data **228** in step **406**. In one example, terrain verification system **234** will conclude from the lack of obstruction from visual evidence that any impediment to safe delivery of cargo **220** would arise from topological anomalies to the soil at destination **308**. Using range data **228** from terrain mapper **238**, terrain verification system **234** can increase or decrease roughness metric **244** accordingly.

(55) As shown in FIG. 5, in step **424** of method **500**, roughness processor **243** compares roughness metric **244** with roughness threshold **236**. Roughness metric **244** at this stage may be calculated based on previous step **406** or previous step **422**. If roughness metric **244** meets or exceeds roughness threshold **236**, roughness processor **243** may deem the surface of destination **308** unsatisfactory and, at step **416**, machine **204** calls an operator for assistance. If roughness metric **244** is below roughness threshold **236**, roughness processor **243** may deem the surface of

destination **308** satisfactory to receive cargo **220** and, at step **410**, causes machine **204** to deposit cargo **220** on the surface of destination **308**.

(56) Those of ordinary skill in the field will appreciate that the principles of this disclosure are not limited to the specific examples discussed or illustrated in the figures. For example, while verification of terrain has been discussed in the context of dropping cargo at a destination site in a worksite, other uses for the process are feasible. The processes could be implemented to evaluate whether a path is sufficiently smooth and level for handling certain kinds of traffic or equipment that may pass over it. Moreover, while the present disclosure addresses particular sequences in FIGS. **4** and **5** for using image data, range data, and inertial data, other sequences of the method steps for cross-checking each of these data sources to deem a drop site to be acceptable are contemplated. In addition, the principles disclosed are not limited to implementation on a semi-autonomous machine in an outdoor worksite. Any machine operating under some level of computer control at worksite inside or outside could benefit from the examples and techniques disclosed and claimed.

INDUSTRIAL APPLICABILITY

(57) The present disclosure provides a terrain verification system associated with a work machine for delivering a pallet to a land surface at a destination. Operating autonomously or semi-autonomously, the work machine includes a range sensor, an image sensor, and an inertial sensor to assess the slope and roughness of the land surface on approach to the destination. Using range data received by the range sensor, such as a LIDAR device, the terrain verification system generates a ground map of the destination. With image analysis of image data received by the image sensor, the system also checks for any obstructions on the land surface. When obstructions are found, the system moves the work machine over the land surface and evaluates pitch and roll data from the inertial sensor to determine the condition of the destination for dropping the pallet.

(58) As noted above with respect to FIGS. **1-5**, a work machine such as machine **204** includes an image sensor **222**, a range sensor **226**, and an inertial sensor **230**. Terrain verification system **234** embedded in hardware and software, typically within machine **204**, processes data from these sensors to determine a roughness metric. In one example, terrain verification system **234** analyzes image data **224** as a further check on the condition of the land surface of the destination following generation of a ground map using a LIDAR device as range sensor **226**. If image data **224** indicates an obstruction, terrain verification system **234** causes machine **204** to drive over the land surface and uses inertial data **232** to determine whether the pallet may be delivered despite the obstruction, such as with light vegetation. In another example, the terrain verification system **234** generates the ground map after identifying an obstruction within image data **224**.

(59) Terrain verification system **234** may be implemented efficiently within control electronics of machine **204**. In some examples, terrain verification system **234** includes a terrain mapper **238** for generating the ground map from range data **228**, an obstruction detector **240** for identifying any obstructions at the destination from image data **224**, and a localization system **242** for evaluating whether an obstruction within image data **224** is an impediment to delivery based on at least pitch and roll data as machine **204** traverses the destination. A roughness processor **243** may calculate a value representative of the slope and roughness of the land surface and compare that value to a predetermined roughness value. Based on the comparison, the work machine may drop the pallet at the destination or alert an operator for help.

(60) In the examples of the present disclosure, terrain verification system **234** helps ensure safe delivery at a drop site by an autonomous or semi-autonomous work machine of cargo **220** that may be fragile or otherwise affected by an irregular land surface. With terrain verification system **234**, an operator of multiple semi-autonomous machines, for instance, can focus on other activities while a machine is sent for delivery to a drop site. If terrain verification system **234** concludes that the drop site is hazardous for delivery due to being uneven, blocked, or rutted, as an example, the work machine will alert the operator for further instructions. If terrain verification system **234**

concludes that the condition of drop site is acceptable after assessing input from various ones of image sensor **222**, range sensor **226**, and inertial sensor **230**, the work machine will proceed with the instructed drop of the pallet **218** with cargo **220** at the destination.

(61) Accordingly, work machines equipped with terrain verification system **234** can decrease risk of damage to cargo **220** when delivered to one of drop points **108**. Loss of material and complications for the work machine with delivery can be avoided, increasing operating efficiency for the autonomous or semi-autonomous machines. Further, the assessment of range data, image data, and inertial (or orientation) data consistent with the present disclosure can enable a work machine to verify the topological condition of a drop site not otherwise apparent to an operator from visual inspection.

(62) Unless explicitly excluded, the use of the singular to describe a component, structure, or operation does not exclude the use of plural such components, structures, or operations or their equivalents. As used herein, the word “or” refers to any possible permutation of a set of items. For example, the phrase “A, B, or C” refers to at least one of A, B, C, or any combination thereof, such as any of A; B; C; A and B; A and C; B and C; A, B, and C; or multiple of any item such as A and A; B, B, and C; A, A, B, C, and C; etc.

(63) Terms of approximation are meant to include ranges of values that do not change the function or result of the disclosed structure or process. For instance, the term “about” generally refers to a range of numeric values that one of skill in the art would consider equivalent to the recited numeric value or having the same function or result. Similarly, the antecedent “substantially” means largely, but not wholly, the same form, manner or degree, and the particular element will have a range of configurations as a person of ordinary skill in the art would consider as having the same function or result. As an example, “substantially parallel” need not be exactly 180 degrees but may also encompass slight variations of a few degrees based on the context.

(64) While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed systems and methods without departing from the spirit and scope of what is disclosed. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. A movable work machine, comprising: a chassis; a range sensor, coupled to the chassis, configured to generate range data indicative of topology of a land surface at a destination of the movable work machine; an image sensor, coupled to the chassis, configured to generate image data indicative of the topology; an inertial sensor, coupled to the chassis, configured to generate orientation data for the movable work machine during traversal of the land surface; and a terrain verification system, comprising: a terrain mapper configured to convert the range data into a ground map of the land surface; an obstruction detector configured to identify one or more obstructions on the land surface from the image data; a localization system configured to evaluate orientation of the movable work machine from at least the orientation data; and a roughness processor configured to determine a roughness metric representative of the topology of the land surface based on at least the ground map and to evaluate the roughness metric for accuracy with respect to the image data, wherein the roughness processor is further configured to compare the roughness metric with a predetermined roughness threshold, and release cargo from the work machine on the land surface at the destination when the roughness metric is less than the predetermined roughness threshold.

2. The movable work machine of claim 1, wherein the roughness processor is further configured to evaluate the roughness metric for accuracy based on the orientation during the traversal of the

movable work machine over the one or more obstructions.

3. The movable work machine of claim 2, wherein the roughness processor is further configured to update the roughness metric using, at least in part, the orientation during the traversal of the movable work machine over the one or more obstructions.

4. The movable work machine of claim 3, wherein the roughness processor is further configured to determine the roughness metric based on the image data.

5. The movable work machine of claim 1, wherein the range sensor is one of a LIDAR device or a stereo camera device configured to provide depth assessment.

6. The movable work machine of claim 1, wherein the roughness processor is further configured to alert an operator when the roughness metric is greater than or equal to the predetermined roughness threshold.

7. A method for verifying terrain at a drop site for a work machine, the method comprising: receiving, by the work machine, an image signal indicative of image data of land surface at the drop site; receiving, by the work machine, a range signal indicative of range data of the land surface; detecting, by the work machine, an obstruction on the land surface from the image data; based at least in part on the detecting the obstruction, causing the work machine to move over the land surface; during movement over the land surface, receiving, by the work machine, inertial data indicative of at least pitch and roll of the work machine; determining a roughness metric for the land surface based at least in part on the pitch and roll of the work machine from the inertial data; comparing the roughness metric with a predetermined roughness threshold; determining the roughness metric to be less than the predetermined roughness threshold; and releasing the payload from the work machine on the land surface at the destination when the roughness metric is less than the predetermined roughness threshold.

8. The method of claim 7, further comprising: determining the roughness metric to be greater than or equal to the predetermined roughness threshold; generating a notification; and sending the notification to an operator of the work machine for assistance.

9. The method of claim 7, further comprising: generating, by a range sensor, the range signal for detecting objects in the environment about the work machine.

10. The method of claim 7, further comprising: prior to detecting the obstruction, converting the range data into a ground map of the land surface; and determining an initial roughness metric based at least in part on the ground map.

11. The method of claim 10, further comprising: determining the initial roughness metric to be less than the predetermined roughness threshold; and releasing cargo from the work machine onto the land surface at the destination.

12. The method of claim 11, wherein determining the roughness metric comprises updating the initial roughness metric based on the pitch and roll of the work machine from the inertial data.

13. The method of claim 11, further comprising: causing the work machine to advance towards the drop site, wherein the receiving the image signal and the receiving the range signal occur during the advance of the work machine towards the drop site.

14. The method of claim 7, further comprising: prior to detecting the obstruction, converting the range data into a ground map of the land surface, wherein the roughness metric is based on the ground map and the pitch and roll of the work machine.

15. A semi-autonomous compact loader, comprising: a chassis; traction devices coupled between the chassis and ground; a work tool coupled to a front of the chassis with respect to a direction of forward travel of the semi-autonomous compact loader, the work tool being configured to enable lifting and lowering a payload; a range sensor configured to generate range data indicative of topology of a land surface at a destination proximate to the semi-autonomous compact loader; an image sensor configured to generate image data indicative of the topology; an inertial sensor configured to generate orientation data for the semi-autonomous compact loader during traversal of the land surface; a terrain mapper configured to convert the range data into a ground map of the

land surface; an obstruction detector configured to identify one or more obstructions on the land surface from image data; a localization system configured to evaluate orientation of the semi-autonomous compact loader from at least the orientation data; and a roughness processor configured to determine a roughness metric representative of the topology of the land surface based on at least the ground map and to evaluate the roughness metric for validity with respect to the image data, wherein the roughness processor is further configured to compare the roughness metric with a predetermined roughness threshold and to release the payload from the semi-autonomous compact loader on the land surface at the destination when the roughness metric is less than the predetermined roughness threshold.

16. The semi-autonomous compact loader of claim 15, wherein the roughness processor is further configured to evaluate the roughness metric for validity based on the orientation during the traversal of the semi-autonomous compact loader over the one or more obstructions.

17. The semi-autonomous compact loader of claim 16, wherein the roughness processor is further configured to refine the roughness metric based on the orientation during the traversal of the semi-autonomous compact loader over the one or more obstructions.

18. The semi-autonomous compact loader of claim 15, wherein the image sensor is one of a high dynamic range (HDR) camera and an ultra-sonic camera.

19. The semi-autonomous compact loader of claim 15, wherein the roughness processor is further configured to alert an operator when the roughness metric is greater than or equal to the predetermined roughness threshold.

20. The semi-autonomous compact loader of claim 15, wherein the range sensor is one of a LIDAR device or a stereo camera device configured to provide depth assessment.
